Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 16: Water: A Precious Resource

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	16 — Water: A Precious Resource
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors are close-but-wrong: surface coverage (71%) mistaken for usable supply, salty ocean water (97%) called fresh, glacier ice treated as the everyday drinking source, recharge confused with withdrawal, paving/deforestation thought to raise the table, falling-table symptoms (deeper wells, costlier pumping) read as causes, drip irrigation accused of starving crops, and the water cycle's constant total misread as water being lost forever. Do not write on this paper — mark answers on your answer sheet.

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1. Of all the water on Earth, about 71% of the planet's surface is covered by water, yet roughly 97% of the total water is in the oceans. A student concludes 'so about 71% of Earth's water is drinkable'. The flaw is that:

(A) 71% surface coverage proves 71% of the water is fresh and drinkable
(B) the 97% ocean figure is wrong; oceans are actually fresh
(C) surface coverage is not the same as usable supply — almost all of that water is salty ocean water, leaving only a tiny fresh fraction
(D) surface coverage and salinity both make all the water drinkable

2. If you imagine all of Earth's water shared into 100 equal buckets, the number of buckets that hold salty ocean water is closest to:

(A) about 71 buckets
(B) about 97 buckets
(C) about 50 buckets
(D) about 3 buckets of salty water and 97 fresh

3. Two friends argue. Riya: 'Glaciers hold a lot of fresh water, so glacier melt is our main daily drinking source.' Sam: 'Most usable fresh water we tap day to day is groundwater and surface water.' Who is right and why?

(A) Riya — glaciers are the everyday tap source for most people

(B) Sam — although glaciers store much fresh water, it is largely locked as ice, so day-to-day usable fresh water comes mainly from groundwater and surface sources

(C) Both are wrong; oceans supply our daily drinking water

(D) Riya — frozen glacier water is the easiest to use directly

4. Rank these from the LARGEST to the smallest store of water on Earth: groundwater, oceans, readily usable river water.

(A) readily usable river water → groundwater → oceans

(B) groundwater → oceans → readily usable river water

(C) oceans → groundwater → readily usable river water

(D) oceans → readily usable river water → groundwater

5. 'There is so much water on Earth that fresh water can never run short.' Which single fact most directly refutes this?

(A) fresh water is being created faster than it is used

(B) rain instantly refills every well the moment it is pumped

(C) the overwhelming majority of Earth's water is salty or frozen, so the usable fresh fraction is genuinely small and can be over-used

(D) oceans can be drunk directly whenever fresh water is low

6. India has roughly 17% of the world's population but only about 4% of its fresh-water resources. The best reading of these two numbers together is that:

(A) India has surplus water and should export it

(B) India has no fresh water of its own

(C) population size has no link to how much water pressure a country feels

(D) India faces water stress — a large share of people must share a much smaller share of fresh water

7. On a winter morning, frost (solid ice crystals) forms on a metal gate, by afternoon it has become liquid droplets, and by evening the gate is dry. The water has passed through which sequence of states?

(A) liquid → solid → vapour

(B) vapour → liquid → solid

(C) solid → liquid → vapour

(D) solid → vapour → liquid

8. Ice cubes, the water in a glass, and the invisible moisture in the air are best described as:

(A) the same substance, water, in three different physical states

(B) three different chemical substances that merely look alike

(C) only two states, since vapour is not really water

(D) water in solid and liquid form, plus steam which is a separate gas

9. A driller reports: 'For the first few metres the soil pores held only air; below a certain depth every pore was full of water.' The depth where the pores first become fully water-filled is the:

(A) water table (B) aquifer
(C) infiltration line (D) saline boundary

10. Two wells are dug side by side into the same ground. Well P is bored into a thick layer of porous sandstone that readily yields water; Well Q hits solid, non-porous granite and stays nearly dry. The sandstone layer is acting as an:

(A) aquifer — a porous rock layer that stores and yields groundwater
(B) water table — the surface of the groundwater
(C) recharge pit — a structure that collects rooftop rain
(D) impermeable seal that blocks all water

11. Water held in the spaces between soil particles and rock below the surface is called groundwater. The single process that puts it there from above is:

(A) evaporation of water from the soil surface
(B) infiltration — water seeping down through the soil
(C) transpiration of water from plant leaves(D) condensation of vapour onto cool rock

12. Over one year a town's aquifer gains 50 units of water by recharge and loses 80 units to pumping. Repeated for several years, the water table will:

(A) fall steadily, because more is withdrawn than is replaced each year
(B) stay level, since recharge and pumping always balance out
(C) rise, because pumping draws fresh water up into the aquifer
(D) fall the first year then recover on its own

13. A region wants to keep its water table steady. Which condition must hold over the long run?

(A) pumping must always exceed recharge by a small amount
(B) the water drawn out roughly equals the water that recharges in
(C) recharge must be zero so the stored water is never disturbed
(D) rainfall must be banned so none of the groundwater escapes

14. A growing city covers most of its open soil and parks with concrete and tarmac. The most direct effect on its groundwater is that:

(A) more rain now soaks in, so the table rises
(B) the rain that lands on concrete becomes salty
(C) the concrete pumps stored water back up to the surface
(D) less rain can infiltrate, so recharge drops and the water table tends to fall

15. Which combination would push a region's water table DOWN the fastest?

(A) good rainfall with many recharge pits and dense forest

(B) scanty rainfall together with rising population, more industry and heavy concretisation

(C) drip irrigation on all farms plus widespread rooftop harvesting

(D) tree planting, leak repair and unpaved open ground

16. Cutting down a forest on a hillside tends to lower the water table downstream. The best causal chain is:

(A) fewer trees → less rain held back and slowed → more runs off and less infiltrates → recharge falls → water table drops

(B) fewer trees → the Sun cools the soil → rain freezes → table drops

(C) fewer trees → rainfall stops entirely → no water at all

(D) fewer trees → rain turns salty → wells become undrinkable

17. After a long, steady monsoon falling on open, vegetated soil, the water table in that area is most likely to:

(A) fall, because rain water mainly evaporates before soaking in

(B) stay frozen until the dry season (C) turn salty from the rain

(D) rise, because sustained infiltration recharges the aquifer

18. Two neighbouring plots get identical rainfall. Plot X is open garden soil; Plot Y is fully tiled. After the monsoon, the water table is higher under:

(A) Plot Y, because tiles trap and store the rain underground

(B) neither — tiling and soil recharge equally

(C) Plot X, because open soil lets rain infiltrate and recharge the groundwater

(D) Plot Y, because gardens waste the rain through plants

19. A village notices its hand-pumps, which once gave water at 8 m depth, now run dry until 18 m. The most direct explanation is that:

(A) the water table has risen, pushing water deeper

(B) the rain has become heavier than usual

(C) the water table has fallen, so water is now reached only at greater depth

(D) the aquifer rock has turned impermeable overnight

20. Which of the following, by itself, would tend to RAISE a depleted water table?

(A) installing more powerful tubewells to pump faster

(B) paving the remaining open ground for parking

(C) diverting all rain straight into a sea-bound drain

(D) building recharge pits so collected rain seeps down into the aquifer

21. An industrial zone keeps drawing groundwater while a new flyover and parking lots cover what was open field. The two effects combine to:

(A) increase withdrawal while cutting recharge, accelerating the fall of the water table

(B) increase recharge while cutting withdrawal, raising the table

(C) leave the water table unchanged because the effects cancel

(D) make the groundwater saline without changing its level

22. As a region's water table keeps falling year after year, which set of consequences is most expected?

(A) wells overflow and pumping becomes free

(B) rainfall automatically increases to compensate

(C) wells and tubewells must be deepened, pumping costs rise, and shallow wells go dry

(D) the soil above turns permanently to ice

23. A farmer's open dug-well went dry this summer for the first time in 20 years, even though the monsoon was normal last year. The most likely chapter-based cause is:

(A) the well water evaporated straight up through the rock

(B) one normal monsoon permanently destroyed the aquifer

(C) long-term over-extraction in the area has lowered the water table below the well's bottom

(D) the well turned salty and so refuses to hold water

24. Within the same country, one district enjoys plentiful water while a neighbouring district faces shortage. This illustrates that fresh water is:

(A) spread perfectly evenly everywhere

(B) distributed unequally — availability varies from place to place

(C) found only in the oceans

(D) always greatest where the population is largest

25. A house leads the rain falling on its roof through a downpipe into a pit filled with gravel and sand so the water slowly seeps into the ground. This setup is:

(A) drip irrigation of the garden (B) flood irrigation of the lawn

(C) rooftop rainwater harvesting feeding a recharge pit

(D) an open ocean desalination plant

26. Why does a gravel-filled recharge pit raise the water table more effectively than letting the same rain run straight into a closed drain?

(A) the pit lets the collected water infiltrate into the aquifer instead of being carried away as runoff

(B) the pit chemically converts rain into groundwater

(C) the drain pumps water upward out of the soil

(D) the pit heats the water so it sinks faster on its own

27. A colony makes rooftop rainwater harvesting common across all its houses. Over a few years its groundwater table most likely:

(A) falls sharply, because harvesting removes water from the ground

(B) becomes salty from collecting roof rain

(C) stays frozen through the harvesting season

(D) rises, because more rain is steered into the ground instead of running off

28. Traditional structures like a bawri (step-well), tanka and johad were built mainly to:

(A) generate electricity from falling water(B) measure the daily rainfall of a region

(C) desalinate sea water for coastal towns

(D) collect, store or recharge rain and groundwater for use in dry periods

29. A farmer switches a field from flood irrigation to drip irrigation. Compared with before, the field now:

(A) uses much less water by delivering it drop by drop to the roots, easing demand on groundwater

(B) uses much more water and lowers the table faster

(C) needs no water at all once drip pipes are laid

(D) makes the groundwater salty as a side effect

30. Among flood irrigation, sprinkler irrigation and drip irrigation, the method that delivers water with the LEAST waste is:

(A) flood irrigation (B) sprinkler irrigation

(C) all three waste equally (D) drip irrigation

31. Besides saving water, drip irrigation also tends to reduce weeds between crop rows because:

(A) the drip water contains a built-in weedkiller

(B) only the crop's root zone is kept moist, so weeds in the dry gaps get little water

(C) the drops flood and wash the weeds away

(D) drip freezes the soil so nothing can grow

32. A market gardener must water 100 potted plants using as little water as possible. The best of these methods is:

(A) a drip line releasing water slowly at each pot's roots

(B) flooding the whole bench so water reaches every pot

(C) an overhead sprinkler running all afternoon

(D) hosing water over the leaves from above

33. During a drought a row of vegetable plants is left without enough water for days. The most likely visible result is that the plants:

(A) grow taller and greener than before (B) turn directly into water vapour

(C) begin flowering far more heavily

(D) wilt — leaves and stems droop and yields fall

34. A farmer reports that a prolonged water shortage cut this year's wheat harvest sharply. The chapter explanation is that:

(A) less water makes crops grow faster and yield more

(B) plants need water to grow and produce, so a shortage stunts growth and lowers the crop

(C) wheat does not need water at all once sown

(D) the shortage made the wheat plants saltier, not smaller

35. World Water Day, marked each year to remind people to value and conserve water, falls on:

(A) 5 June (B) 22 April

(C) 14 November (D) 22 March

36. A poster for World Water Day declares 'every living thing needs water'. Which statement best supports this theme within the chapter?

(A) only crops need water; animals can live without it

(B) humans, animals and plants all depend on water for their life processes, so water must be used carefully

(C) water matters only to people living in cities

(D) since oceans are huge, no living thing ever lacks water

37. A town's water-use survey shows: drinking and cooking 8%, washing and bathing 30%, irrigation of nearby farms 55%, industry 7%. To cut groundwater stress most, the town should first target:

(A) drinking water, the smallest use (B) industry, since 7% is the biggest share

(C) irrigation, since it is by far the largest use and can switch to water-saving drip methods

(D) nothing, since none of the uses can be reduced

38. Reservoir readings over four years: Year 1 full, Year 2 down a little, Year 3 down more, Year 4 lowest. Rainfall was normal each year but the town added many new tubewells. The pattern best indicates:

(A) withdrawals have outpaced recharge, steadily depleting the stored water

(B) recharge has outpaced withdrawal, so storage is rising

(C) rainfall failed completely every year

(D) the falling level proves the water turned to vapour

39. A school audit finds a single tap dripping about 30 litres a day across the whole campus's many taps. The most sensible response is to:

(A) mend the leaks promptly, since many small drips add up to a large daily loss

(B) ignore them, as one drip is far too small to matter

(C) open every tap fully so it cannot drip (D) leave a bucket under each tap forever

40. Which set of actions ALL increase groundwater recharge?

(A) paving open ground, flood irrigation, draining rain to the sea

(B) rooftop rainwater harvesting, building recharge pits, keeping ground open and vegetated

(C) deep tubewell pumping, concreting parks, cutting forests

(D) leaving taps running, long showers, filling more swimming pools

41. A city planner must choose ONE measure to most reliably slow a falling water table. The best choice is:

(A) sanction more bore wells to share the demand

(B) tile over the city's open grounds for cleanliness

(C) route all storm water out of the city as fast as possible

(D) require rainwater-harvesting recharge structures on new buildings

42. Why can two villages receiving the SAME rainfall still end up with very different water tables after some years?

(A) rainfall is the only factor, so their tables must be identical

(B) the village with more concrete always has the higher table

(C) they may differ in how much they pump out and how much rain they let infiltrate versus run off

(D) the deeper a village digs its wells, the higher its table rises

43. A correct natural-recharge chain for groundwater is:

(A) rain falls → runs off concrete → into drains → water table rises

(B) groundwater is pumped out → soil dries → water table rises

(C) sea water evaporates → blows inland → water table rises directly

(D) rain falls → soaks into open soil → percolates down → reaches the aquifer → water table rises

44. Coastal areas that over-pump groundwater can find their wells turning salty. Within the chapter's logic, this is because:

(A) pumping chemically converts fresh water into salt

(B) lowering the fresh groundwater lets nearby salty water move in toward the wells

(C) rain over the coast is naturally salty and fills the wells

(D) concrete near the coast manufactures salt in the soil

45. A town runs all four measures together: rooftop harvesting, drip irrigation on nearby farms, repairing leaking mains, and planting trees on open ground. Over several years its water table most likely:

(A) falls faster than before, because more structures use more water

(B) rises, because recharge improves while wasteful withdrawals fall

(C) turns salty from the combined measures

(D) stays exactly the same forever, unaffected by any measure

46. Which statement correctly compares the water amounts on Earth?

(A) readily usable fresh water exceeds all the ocean water

(B) glacier ice is the source most of us drink from each day

(C) salty ocean water vastly exceeds the fresh water we can readily use

(D) all of Earth's water is fresh and drinkable

47. A student claims, 'Because water vapour goes up into the air, water is constantly being lost from Earth forever.' The correct view is that:

(A) water changes form and place but the total on Earth stays roughly the same — vapour later returns as rain

(B) water vapour escapes to space, so Earth slowly dries out

(C) vapour is destroyed in the air, reducing total water

(D) vapour becomes a new substance and never returns as water

48. 'Drip irrigation is bad because it gives plants very little water.' The best correction is:

(A) true — drip starves crops and should be avoided

(B) drip actually floods the field, so the claim is backwards

(C) drip gives each plant enough at its roots while wasting little elsewhere — it saves water without starving the crop

(D) drip is only for ornamental plants, never food crops

49. Why is rooftop rainwater harvesting often promoted in cities specifically?

(A) cities receive far more rain than any village

(B) city roofs and pavements shed huge volumes of rain that would otherwise run off without recharging the ground

(C) city groundwater can never be depleted anyway

(D) harvesting works only where there is no soil at all

50. A region's water table has been falling for a decade. Which response treats the CAUSE rather than just the symptom?

(A) drilling every well deeper to keep reaching the dropping water

(B) buying more powerful pumps to lift the lower water

(C) cutting over-extraction and boosting recharge so withdrawals no longer exceed inflow

(D) importing bottled water for daily drinking

51. Which observation would be the clearest evidence that a recharge programme is WORKING in an area?

(A) the nearby ocean becomes less salty

(B) wells that had dried up start giving water again at shallower depths over a few seasons

(C) rainfall during the monsoon suddenly doubles

(D) the rivers stop flowing entirely

52. A farmer floods a whole field to water a few rows of crops. From a water-conservation view, the main problem is that:

(A) flooding gives the crop too little water to survive

(B) flooding makes the groundwater rise too fast and harm the crop

(C) flooded fields cannot grow any crop at all

(D) much of the water runs off or evaporates from bare ground instead of reaching the crop roots

53. A village builds a johad (earthen check-dam) across a seasonal stream. The main way it helps the water table is by:

(A) pumping groundwater up to the surface for use

(B) sealing the streambed so no water can ever soak in

(C) evaporating the stored water quickly into the air

(D) holding back rain water so it has time to seep into the ground and recharge the aquifer

54. Sea water is unsuitable for drinking or watering most crops chiefly because it is:

(A) salty, which harms the body and most plants

(B) too cold to drink safely (C) permanently frozen into ice

(D) too deep in the ocean to reach

55. A planner says: 'If we double the city's tubewells, we double our water supply for good.' The chapter-based flaw is that:

(A) more tubewells create new water out of nothing

(B) the aquifer is unlimited, so the plan is perfectly fine

(C) tubewells add rain to the sky to refill themselves

(D) tubewells only withdraw from a limited aquifer; pumping faster than recharge soon lowers the table and dries wells

56. Which statement best captures the chapter's central message that 'water is a precious resource'?

(A) water is unlimited, so saving it is pointless

(B) only farmers, not households, need to conserve water

(C) the sea will always supply enough drinking water

(D) usable fresh water is genuinely scarce, so we must use it wisely and manage it carefully

57. A study compares two farms in the same dry district. Farm A uses flood irrigation and its well dries by April; Farm B uses drip irrigation and its well lasts till June. The fairest conclusion is that:

(A) flood irrigation conserves more groundwater than drip

(B) irrigation method has no effect on how long a well lasts

(C) drip irrigation's lower water use helped Farm B's well last longer

(D) Farm B simply received more rain than Farm A

58. Why does 'every living thing needs water' make conserving fresh water everyone's responsibility, not just farmers'?

(A) only crops use water, so only farmers must save it

(B) all people, animals and plants depend on the same scarce fresh supply, so any wasteful use anywhere strains it

(C) households use no water worth saving

(D) wild animals and plants get their water from the salty sea

59. A region installs both rooftop harvesting (more recharge) and drip irrigation (less withdrawal). If recharge now exceeds withdrawal, the water table over time will:

(A) rise, because more water enters the aquifer than leaves it

(B) fall, because two measures together drain the aquifer

(C) stay flat, since recharge and withdrawal must always match

(D) rise only if the measures are removed afterwards

60. Concretising a floodplain that used to soak up monsoon water typically causes which TWIN effect?

(A) more surface flooding during rain and less groundwater recharge afterwards

(B) less flooding and more recharge (C) no flooding and no change in recharge

(D) more recharge but also more flooding

Solutions — Science (Class 7), Chapter 16: Water: A Precious Resource — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	A	31	B	41	D	51	B
2	B	12	A	22	C	32	A	42	C	52	D
3	B	13	B	23	C	33	D	43	D	53	D
4	C	14	D	24	B	34	B	44	B	54	A
5	C	15	B	25	C	35	D	45	B	55	D
6	D	16	A	26	A	36	B	46	C	56	D
7	C	17	D	27	D	37	C	47	A	57	C
8	A	18	C	28	D	38	A	48	C	58	B
9	A	19	C	29	A	39	A	49	B	59	A
10	A	20	D	30	D	40	B	50	C	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The 71% describes how much of the surface is covered, not how much of the water is usable. Of the total water, about 97% is saline ocean water and most of the remaining fresh water is locked in ice or deep underground, so only a tiny slice is readily usable. Confusing area coverage with usable supply, or calling ocean water fresh, is the error.

2. (B) About 97% of all water is in the oceans as salty water, so out of 100 buckets roughly 97 are saltwater. The 71 figure is surface coverage, not a share of the water; 50 is a guess; and the 'mostly fresh' version reverses the true ratio. So about 97 buckets is correct.

3. (B) A large share of fresh water is indeed frozen in glaciers and ice caps, but that ice is not readily available for daily use. The fresh water we actually draw on day to day is groundwater and surface water. Sam is right; ocean water is salty, not a drinking source.

4. (C) Oceans dwarf everything, holding about 97% of all water. Of the small fresh fraction, far more sits underground as groundwater than flows in rivers at any moment. So the order is oceans, then groundwater, then the small readily usable river water. The other orders misrank these.

5. (C) The total is huge, but most of it is salty ocean water or frozen ice — neither is readily usable. The small usable fresh fraction can be depleted faster than it is recharged, so shortages are real. Claiming fresh water self-creates, that wells refill instantly, or that oceans are drinkable are all false.

6. (D) A big population share (17%) drawing on a small water share (4%) means each person, on average, has little to go round — that is water stress. It is not surplus, it is not zero, and population clearly does affect per-person water pressure. So 'water stress' is the correct interpretation.

7. (C) Frost is solid ice; melting in the afternoon makes liquid droplets; drying turns the liquid into water vapour that leaves the gate. So the order is solid, then liquid, then vapour. The other sequences put the states in the wrong order for what is observed.

8. (A) Ice, liquid water and water vapour are all the same substance — water — differing only in physical state (solid, liquid, gas). They are not different chemicals, vapour is genuinely water, and the gaseous water in air is water vapour, not a separate gas. So 'same substance, three states' is correct.

9. (A) The water table is the upper level below which the ground is saturated — pores fully filled with water. The aquifer is the water-bearing rock layer itself, not a depth marker; 'infiltration line' is not a term; and saline boundary refers to salt water, not the saturation level. So it is the water table.

10. (A) An aquifer is a layer of porous rock or gravel that holds and gives up water, which is exactly what the sandstone does. The water table is a level, not a rock layer; a recharge pit is a man-made pit; and the granite is the impermeable layer, not the water-yielding one. So the sandstone is an aquifer.

11. (B) Groundwater is recharged when surface water seeps downward through the soil — that downward seeping is infiltration. Evaporation and transpiration move water upward into the air, and condensation forms liquid from vapour rather than driving water underground. So infiltration is the process.

12. (A) Each year the aquifer loses 30 units net (80 out, 50 in), so the stored water — and hence the water table — keeps dropping year after year. Pumping removes water rather than adding it, and there is no self-recovery while withdrawal exceeds recharge. So the table falls steadily.

13. (B) A steady water table needs a balance: withdrawals must be matched by recharge. If pumping exceeds recharge the table falls; zero recharge would let pumping empty the aquifer; and banning rainfall removes the very source of recharge. So 'draw out ≈ recharge in' is the condition.

14. (D) Concrete and tarmac are nearly impermeable, so rain runs off into drains instead of soaking in. Reduced infiltration means less recharge, so the water table tends to fall. Paving cannot increase soaking, does not make rain salty, and does not pump water up. So recharge drops and the table falls.

15. (B) Falling water tables come from low recharge plus high withdrawal. Scanty rain (little recharge) combined with more people, industry and paving (high demand and even less soaking) is the worst case. The other options all boost recharge or cut waste, which would raise or steady the table. So the first combination drops it fastest.

16. (A) Forests slow runoff and let rain soak in, recharging groundwater. Remove the trees and more water runs off quickly while less infiltrates, so recharge falls and the table drops. Deforestation does not cool the soil, halt rainfall, or salt the rain. So the first chain is correct.

17. (D) Steady rain on open soil infiltrates over many days, adding water to the aquifer faster than it leaves, so the table rises. Most of a long soaking rain seeps in rather than wholly evaporating; groundwater does not freeze in this context; and rain does not salt the aquifer. So the table rises.

18. (C) Open soil allows rain to soak in and recharge the aquifer, while tiles shed rain to drains with little infiltration. So the table rises more under Plot X. Tiles do not trap water underground, the two surfaces do not behave the same, and a garden's modest plant use is far outweighed by the recharge it allows. So Plot X is higher.

19. (C) If water that used to appear at 8 m now needs 18 m, the saturated level — the water table — has dropped. A rising table would bring water nearer the surface; heavier rain would raise, not lower, the table; and aquifer rock does not suddenly turn solid. So the water table has fallen.

20. (D) Recharge pits guide rain into the ground, adding to the aquifer and lifting the table. More powerful pumps and extra paving both worsen depletion, and routing rain straight to the sea wastes the recharge entirely. So recharge pits raise the table.

21. (A) Heavy industrial pumping raises withdrawal, and paving over open field cuts infiltration and thus recharge. More out and less in together drop the table faster. They do not cancel — both push the same way — and paving does not salinise the water. So the table falls faster.

22. (C) A falling table means water sits deeper, so wells must be drilled deeper and pumps work harder (higher cost), while older shallow wells dry up. Wells do not overflow, rainfall is not forced to rise by depletion, and the soil does not freeze. So deepening wells, rising costs and dry shallow wells is the expected set.

23. (C) A well drying up despite a normal monsoon points to a steadily falling water table from years of over-pumping, dropping the saturated level below the well's floor. Groundwater does not evaporate up through rock, one normal monsoon does not wreck an aquifer, and salinity does not stop a well holding water. So over-extraction is the cause.

24. (B) Differing availability between neighbouring districts shows water is unequally distributed across regions. It is not even everywhere, fresh water is not only in oceans (which are salty), and abundance does not track population — often the most populated places are the most stressed. So unequal distribution is the point.

25. (C) Collecting roof rain and letting it seep down through a gravel pit to recharge groundwater is rooftop rainwater harvesting via a recharge pit. Drip and flood irrigation are crop-watering methods, and desalination removes salt from sea water — none describe roof-to-pit recharge. So it is rooftop rainwater harvesting.

26. (A) A recharge pit holds rain in contact with permeable ground so it soaks down and refills the aquifer; a closed drain simply removes the water as runoff, giving no recharge. Pits do not chemically make groundwater, drains do not pump water up, and heating is not how the pit works. So infiltration versus runoff is the reason.

27. (D) Directing roof rain into the ground increases recharge, so over time the table rises. Harvesting adds water rather than removing it, roof rain is fresh (not salty), and groundwater does not freeze here. So the table rises.

28. (D) Bawris, tankas and johads are traditional water-harvesting and storage works that capture rain or groundwater for the dry season. They are not power plants, rain gauges, or desalination units. So storing and recharging water is their purpose.

29. (A) Drip irrigation feeds water directly to each plant's root zone, cutting the large losses of flood irrigation, so less groundwater is needed. It does not increase use, crops still need water, and drip does not salinise groundwater. So it uses much less water and eases demand.

30. (D) Drip targets the root zone directly, losing little to runoff or evaporation; sprinklers lose more to spray and evaporation; flooding loses the most by spreading water over the whole field. The three are not equal. So drip irrigation wastes the least.

31. (B) Drip wets only narrow strips at the crop roots, leaving the spaces between rows dry, so weeds there are starved of water. There is no weedkiller in plain water, drip does not flood, and it does not freeze the soil. So the dry gaps suppressing weeds is the reason.

32. (A) Drip delivers water straight to each pot's roots with little loss, the most water-efficient choice. Flooding, all-afternoon sprinkling, and hosing the leaves all lose far more to runoff and evaporation. So the drip line is best.

33. (D) Without enough water, plant cells lose turgidity, so leaves and stems droop (wilting) and crop yields drop. Plants do not grow lusher when starved of water, do not vaporise, and do not flower more. So wilting and reduced yield is the result.

34. (B) Water is essential for plant growth and grain filling; a shortage stunts the crop and reduces yield. Less water does not speed growth, wheat does need water throughout, and a shortage reduces the crop rather than just changing its salt content. So shortage lowering the harvest is correct.

35. (D) World Water Day is observed on 22 March. 5 June is World Environment Day, 22 April is Earth Day, and 14 November is Children's Day in India — none is World Water Day. So 22 March is correct.

36. (B) The chapter stresses that all living things — people, animals and plants — depend on water, so the small usable fresh supply must be conserved. Animals do need water, water matters everywhere not just cities, and the huge salty oceans do not guarantee any creature usable fresh water. So the all-life-depends statement fits.

37. (C) Irrigation at 55% is by far the largest draw, so making it efficient (e.g. drip) yields the biggest saving. Drinking (8%) is small and essential, industry is only 7% (not the biggest), and uses can be reduced. So targeting irrigation gives the largest cut.

38. (A) A steady year-on-year fall under normal rainfall, alongside many new tubewells, shows water is being taken out faster than it comes in — depletion. Storage is clearly falling not rising, rainfall was normal not absent, and evaporation does not explain a multi-year withdrawal-driven decline. So withdrawals outpacing recharge fits.

39. (A) Although one drip seems tiny, many taps each losing 30 litres a day total a large waste, so prompt repair is the sensible fix. Ignoring them lets the loss continue, opening taps fully wastes far more, and permanent buckets do not stop the loss. So mend the leaks.

40. (B) Roof harvesting, recharge pits and keeping ground open and planted all steer more water into the aquifer, raising recharge. The other sets either block infiltration (paving, concreting), remove water (pumping, drains), or simply waste it (running taps, showers, pools). So the first set increases recharge.

41. (D) Mandating recharge structures puts rain back into the ground, directly countering the fall. More bore wells increase withdrawal, tiling over ground cuts infiltration, and rushing storm water out wastes recharge. So requiring recharge structures is the best single measure.

42. (C) Equal rain does not fix the outcome; what differs is withdrawal (pumping) and recharge (infiltration vs runoff, affected by paving and vegetation). More concrete lowers, not raises, a table, and digging deeper wells does not lift the table. So differing pumping and infiltration explain it.

43. (D) Natural recharge is rain infiltrating open soil, percolating down to the aquifer and lifting the table. Runoff to drains gives no recharge, pumping lowers the table, and evaporated sea water must fall and infiltrate before it could recharge — it does not raise the table 'directly'. So the first chain is correct.

44. (B) Heavy pumping drops the fresh groundwater level, allowing adjacent salty water to encroach toward the wells, making them salty. Pumping does not chemically create salt, rain is fresh not salty, and concrete does not make salt. So intruding salt water after over-pumping is the cause.

45. (B) Harvesting and tree-planted open ground boost recharge, while drip and leak repair cut waste — more in, less out, so the table rises over time. These measures save water rather than using more, do not salinise it, and clearly do affect the table. So it rises.

46. (C) Oceans hold about 97% of the water as salt water, far more than the tiny readily usable fresh fraction. Usable fresh water does not exceed ocean water, glacier ice (frozen) is not our everyday drinking source, and not all water is fresh. So salty ocean water vastly exceeding usable fresh water is correct.

47. (A) Water in the cycle changes state and location but is not lost: vapour later condenses and falls back as rain, keeping the total roughly constant. It does not escape to space in any meaningful amount, is not destroyed, and remains water throughout. So 'changes form and place, total constant' is correct.

48. (C) Drip supplies water precisely to the root zone in adequate amounts while cutting losses between rows, so crops are not starved — water is simply not wasted. It does not flood the field and is widely used for food crops. So the saving-without-starving correction is right.

49. (B) Cities are heavily paved, so most rain runs off rather than soaking in; harvesting captures that runoff and recharges the ground, which is why it is pushed in cities. Cities do not necessarily get more rain, their groundwater certainly can deplete, and harvesting is not limited to soil-less places. So capturing urban runoff is the reason.

50. (C) The cause of a falling table is withdrawals exceeding recharge; reducing pumping and increasing recharge fixes that balance. Deeper wells and stronger pumps merely chase the falling level (treating symptoms), and bottled water sidesteps the aquifer entirely. So cutting extraction and boosting recharge addresses the cause.

51. (B) If recharge succeeds, the water table rises, so previously dry wells yield water at shallower depths again — direct evidence. Ocean salinity, doubled rainfall, and rivers stopping are unrelated to the recharge effort. So the wells recovering at shallower depth is the clearest sign.

52. (D) Flooding spreads water over the whole field, so a lot is lost to runoff and evaporation from bare soil rather than feeding the roots — wasteful. It does not under-water the crop, the issue is waste not a too-fast rise, and crops do grow with flooding (just inefficiently). So the loss of water off the crop is the problem.

53. (D) A johad ponds the stream water, giving it time to infiltrate and recharge the aquifer, raising the table. It does not pump water up, it encourages (not blocks) seepage, and quick evaporation would defeat its purpose. So holding water for infiltration is how it helps.

54. (A) Sea water's high salt content makes it unfit to drink and damaging to most crops. It is not the temperature, it is not frozen, and depth is not why it is unusable — salinity is. So 'salty' is the reason.

55. (D) Tubewells draw on a finite aquifer; doubling them speeds withdrawal beyond recharge, lowering the table until wells fail — no lasting gain. They do not create water, the aquifer is not unlimited, and they do not generate rain. So the limited-aquifer point is the flaw.

56. (D) Despite Earth's vast water, the usable fresh share is small and easily over-used, so wise use and careful management matter. Water is not unlimited, conservation is everyone's job, and the salty sea cannot supply drinking water. So the scarcity-and-management message is central.

57. (C) Same district means similar rainfall, so the difference is the method: drip uses less water, easing draw on the well so it lasts longer. Flood irrigation uses more (not less), method clearly matters here, and they share the district's rainfall. So drip's lower use explains Farm B's longer-lasting well.

58. (B) Because every living thing draws on the limited fresh supply, wasteful use by anyone — homes, industry or farms — adds strain, making conservation a shared duty. Crops are not the only users, households use significant water, and wildlife relies on fresh water not the salty sea. So shared dependence makes it everyone's job.

59. (A) When recharge exceeds withdrawal, the aquifer gains water and the table rises. Both measures push toward that balance, not toward draining; recharge and withdrawal do not have to match; and removing the measures would reverse the gain. So the table rises.

60. (A) A concreted floodplain cannot absorb rain, so water pools and floods the surface during downpours while almost nothing soaks in to recharge groundwater — a twin loss. It does not reduce flooding, it does change recharge (lowering it), and it cannot raise recharge while paved. So more flooding with less recharge is correct.